Exp. No: 6 Capacitance of a Parallel Plate Capacitor in Different Dielectric Media

Aim:

To write program for capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constants using MATLAB.

Objectives:

1. To compute the capacitance of a parallel plate capacitor for varying plate area, separation and dielectric constant.

2. To compare the capacitance obtained across three different dielectric mediums.

3. To validate the simulated results using Poisson's and Laplace's equation based theoretical values.

Test case (i) Determination and plotting of capacitance of parallel plate capacitor for given area and dielectric thickness

```
%% Clean workspace

clearvars;

close all;

clc;


%% Constants & User Inputs

eps_0 = 8.854e-12;   % Permittivity of free space (F/m)


area = input('Enter the area value=');

thickness = input('Enter the thickness value=');
```

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```
%% Analytical Capacitance Calculation
cap_analytical = (eps_0 * area) / thickness;

disp('The capacitance is');
disp(cap_analytical);

%% Capacitance vs. Separation Curve
d_range = linspace(1e-3, 10e-3, 50);          % Separation range from 1 mm to 10 mm
cap_range = (eps_0 * area) ./ d_range;

figure('Color', 'w');
plot(d_range * 1e3, cap_range, 'b-', 'LineWidth', 2);
grid on;
xlabel('Plate separation d (mm)');
ylabel('Capacitance (F)');
title('Parallel plate capacitance versus separation');

%% 1D Finite-Difference Laplace Solver
num_nodes = 51;
V_boundary = 100;
```

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```
max_iter = 20000;

% Initialize potential grid with boundary conditions (V(1) = 0, V(end) = V0)
pot = zeros(1, num_nodes);
pot(end) = V_boundary;

% Iterative finite-difference update (d^2V/dx^2 = 0)
for k = 1:max_iter
    pot(2:end-1) = 0.5 * (pot(1:end-2) + pot(3:end));
end

%% Numerical Capacitance Extraction
pos_grid = linspace(0, thickness, num_nodes);
E_field = -gradient(pot, pos_grid);

% C = Q / V0 where Q = eps_0 * mean(|E|) * Area
cap_laplace = (eps_0 * area * mean(abs(E_field))) / V_boundary;

fprintf('Capacitance from Laplace solution = %.4e F\n', cap_laplace);
```

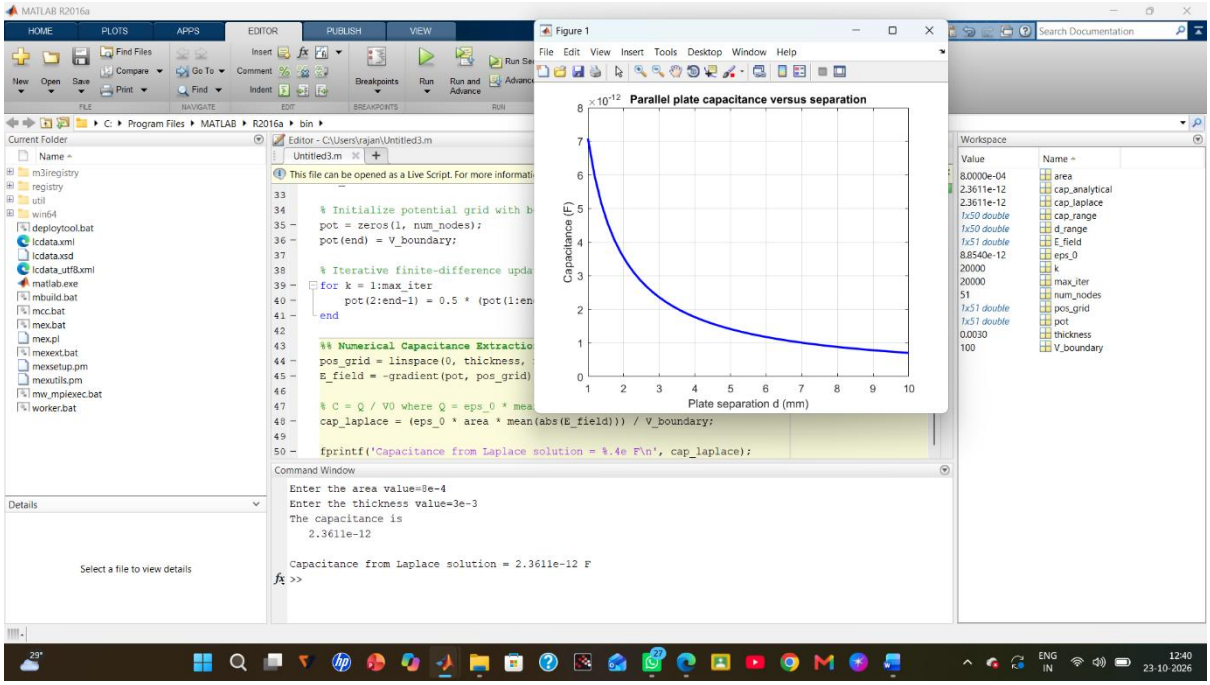
Output:- Enter the area value = 8e-4

Enter the thickness value = 3e-3

The capacitance is 2.36×10^{-12} F

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Capacitance from Laplace solution = 2.3611e-12 F



Test case (ii) Determination of capacitance of parallel plate capacitor with three different dielectric layers

%% User Inputs

plate_area = input('Enter the area value=');

t1 = input('Enter thickness d1=');

t2 = input('Enter thickness d2=');

t3 = input('Enter thickness d3=');

eps_r1 = input('Enter dielectric constant er1=');

eps_r2 = input('Enter dielectric constant er2=');

eps_r3 = input('Enter dielectric constant er3=');

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```
%% Equivalent Capacitance Calculation (Series Dielectric Slabs)
% C = (eps_0 * A) / sum(d_i / eps_ri)
effective_d = (t1 / eps_r1) + (t2 / eps_r2) + (t3 / eps_r3);
total_capacitance = (eps_0 * plate_area) / effective_d;

%% Display Results
disp('The capacitance is');
disp(total_capacitance);
```

OUTPUT:

Enter the area value = 8e-4

Enter thickness d1 = 1e-3

Enter thickness d2 = 3e-3

Enter thickness d3 = 2e-3

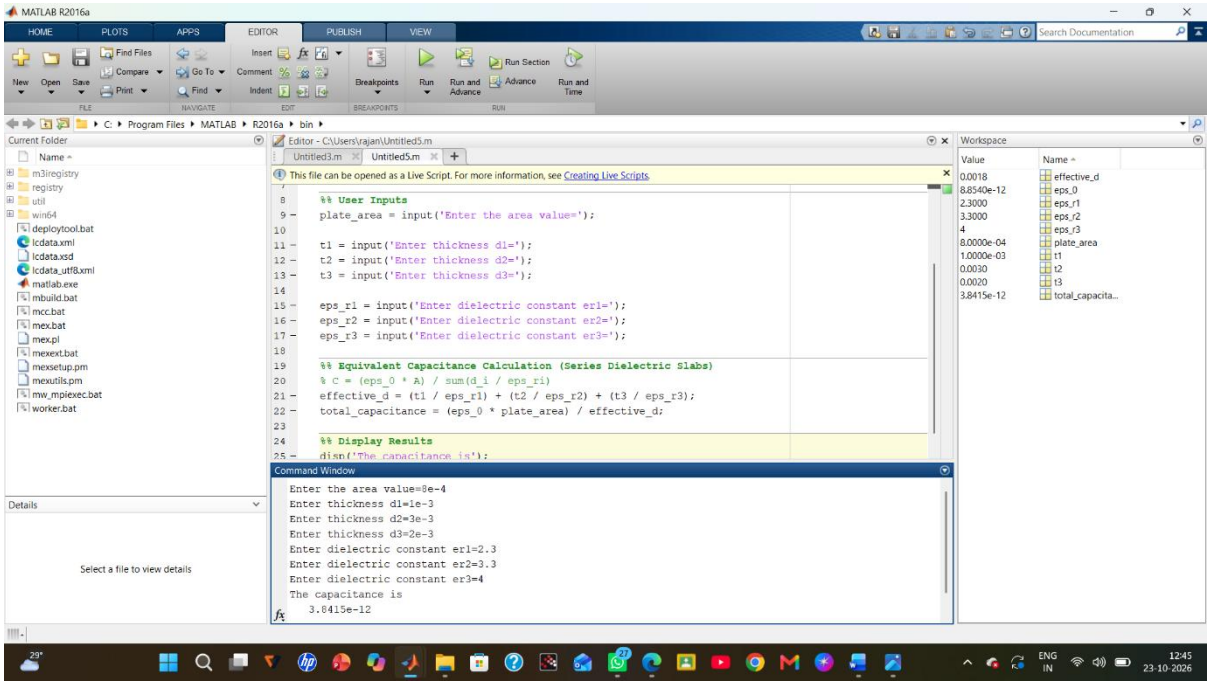
Enter dielectric constant er1 = 2.3

Enter dielectric constant er2 = 3.3

Enter dielectric constant er3 = 4

The capacitance is 3.84×10^{-12} F

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Test case (iii) Determination and plotting of capacitance variation with dielectric constant (Air)

%% Reset Environment

clearvars;

close all;

clc;

%% Physical Constants & Plate Geometry

eps_0 = 8.854e-12; % Permittivity of free space (F/m)

plate_area = 1e-4; % Plate area (m^2)

separation = 1e-3; % Plate distance (m)

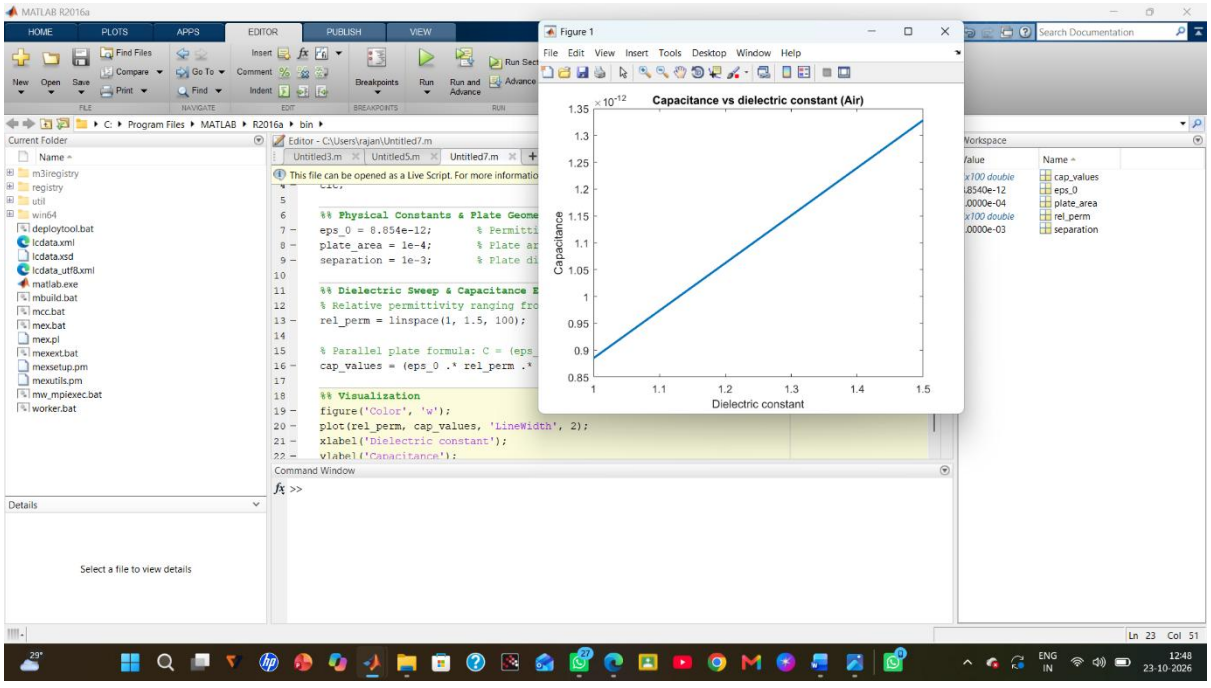
%% Dielectric Sweep & Capacitance Evaluation

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```
% Relative permittivity ranging from 1.0 to 1.5 across 100 sample points
rel_perm = linspace(1, 1.5, 100);

% Parallel plate formula: C = (eps_0 * eps_r * A) / d
cap_values = (eps_0 .* rel_perm .* plate_area) ./ separation;

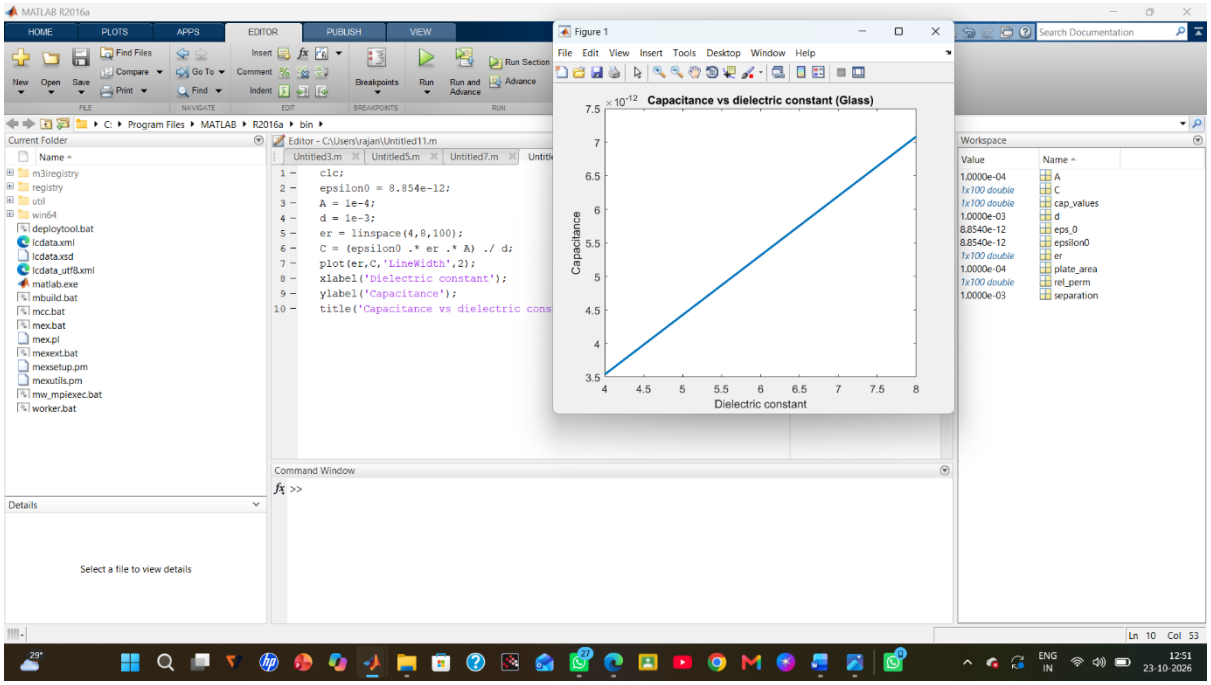
%% Visualization
figure('Color', 'w');
plot(rel_perm, cap_values, 'LineWidth', 2);
xlabel('Dielectric constant');
ylabel('Capacitance');
title('Capacitance vs dielectric constant (Air)');
```



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Test case (iv) Determination and plotting of capacitance variation with dielectric constant (Glass)

```
clc;
epsilon0 = 8.854e-12;
A = 1e-4;
d = 1e-3;
er = linspace(4,8,100);
C = (epsilon0 .* er .* A) ./ d;
plot(er,C,'LineWidth',2);
xlabel('Dielectric constant');
ylabel('Capacitance');
title('Capacitance vs dielectric constant (Glass)');
```



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S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Set capacitor geometry and material properties	Set capacitor geometry and material properties	Correctly enters plate area, separation and relative permittivity with SI units.	Inputs are correct with a minor unit/property issue.	Several inputs need correction.	Geometry or dielectric inputs are invalid	25	2.5	
2	Calculate capacitanc e in MATLAB	Calculate capacitance in MATLAB	Uses the correct capacitance relation and calculates values for all specified media.	Calculat ions are mostly correct with minor errors.	Only some cases are calculated correctly	Capacitance calculations are incorrect.	25	2.5	
3	Compare dielectric and geometry effects	Compare dielectric and geometry effects	Correctly evaluates trends versus permittivity/ area/ separation and validates the model.	Trend and validati on are mostly correct with minor omissio ns.	Trend discussion or validation is incomplete	Incorrectly interprets parameter effects.	30	3	
4	Document engineerin g observatio ns	Document engineering observation s	Provides a complete table/plot with units and explains dielectric influence.	Docume ntation and explanat ion are mostly complet e	Some labels, units or explanations are missing	Results are incomplete or unexplained	20	2	
Tota l							100	10	